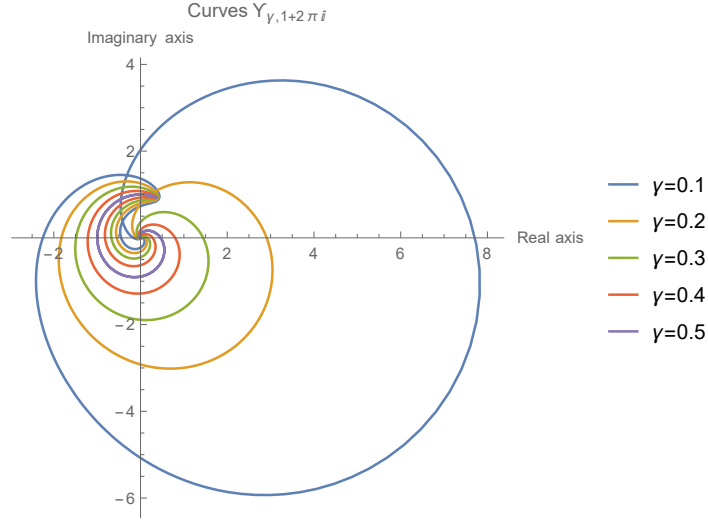


Figure 6:



It seems natural to introduce the subspaces $H_p^{(n)}$ for $1 \leq p \leq \infty$, and consider the generalized Stieltjes operator $\mathcal{S}_{\beta, \mu}$ given by

$$\mathcal{S}_{\beta, \mu} F(z) := z^{\mu-\beta} \int_0^\infty \frac{s^{\beta-1}}{(z+s)^\mu} F(s) ds, \quad z \in \mathbb{C}^+,$$

on the spaces $H_p^{(n)}$ for $0 < \beta - 1/p < \mu$. This will be the focus of a forthcoming paper.

8.3 Open questions

In this paper we have presented a complete study of generalized Stieltjes operator on fractional Sobolev-Lebesgue spaces. However some of these results might be improved.

- (i) The case $p = \infty$ in the Sobolev-Lebesgue spaces $\mathcal{T}_\infty^{(\alpha)}(t^\alpha)$ is commented in Remark 3.1. It seems natural to conjecture that the operator $\mathcal{S}_{\beta, \mu}$ is bounded on these spaces $\mathcal{T}_\infty^{(\alpha)}(t^\alpha)$, $\|\mathcal{S}_{\beta, \mu}\| = B(\beta, \mu - \beta)$, for $0 < \beta < \mu$ and $\sigma(\mathcal{S}_{\beta, \mu}) = \Upsilon_{\beta, \mu} \cup \{0\}$.
- (ii) In Theorem 3.2 we have proved a Hölder inequality in $\mathcal{T}_p^{(\alpha)}(t^\alpha)$ for conjugate exponents $p, p' > 1$ and $\alpha \geq 0$. As in the classical case, we may conjecture that given $f \in \mathcal{T}_p^{(\alpha)}(t^\alpha)$ and $g \in \mathcal{T}_q^{(\alpha)}(t^\alpha)$ then $fg \in \mathcal{T}_r^{(\alpha)}(t^\alpha)$ and

$$\|fg\|_{\alpha, r} \leq C_\alpha \|f\|_{\alpha, p} \|g\|_{\alpha, q},$$

where C_α is a positive constant and $r, p, q \in [1, \infty]$ such that $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$. Note that for $\alpha \in \mathbb{N}$, the statement holds in a straightforward way from the usual Leibniz formula and Hölder inequality.